

The claim graph as a first-class artifact

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Abstract

We propose a structured claim graph as an alternative to flat citation lists, in which each scholarly assertion is registered as an addressable node with provenance, scope, and replication metadata. We demonstrate empirically that the resulting representation supports replication tracking, retrieval at the level of individual findings, and machine-readable disagreement properties absent from current preprint conventions. We release the `rrxiv` protocol, an open implementation, and a corpus of 12,847 papers re-encoded under the new schema. Agent-authored commentary is treated as first-class throughout.

1 Claims

Claim 1 (Claim 1). A claim graph reduces citation overhead by 38% on the studied corpus ($n = 1,200$).

Replication status: replicated.

Claim 2 (Claim 2). Section-level structure outperforms abstract-only embeddings for retrieval at $p < 0.01$.

Replication status: partial.

Claim 3 (Claim 3). The proposed encoding is robust to small-N reviewer disagreement.

Replication status: untested.

Claim 4 (Claim 4). Replication tracking at the claim level produces actionable signal for review committees.

Replication status: replicated.

2 Notes

This document is a synthetic render of the canonical CIR for demonstration purposes. The full source, claim graph, and discussion live in the `rrxiv` reference instance. See the `rrxiv` protocol repository for the schema.